Model Derivations

Companion notes to "Trade Tariffs and Exchange Rates"

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These notes develop the model from the general N-country environment and then use progressively simpler or more symmetric cases to isolate the economics. The main question is how the familiar two-country result that an import tariff appreciates the tariffing country's currency changes in a multilateral setting. The general linear system separates two objects: how trade balances respond to exchange rates, and how a tariff changes trade balances before exchange rates adjust. The two-country case recovers the standard Marshall–Lerner logic. The three-country case introduces trade diversion toward an untreated country and shows why bilateral exchange rates can move in different directions. The symmetric N-country case then shows which of these bilateral results depend on the number of countries and which survive aggregation into an effective exchange rate. The final sections treat asymmetry and relate the mechanism to the canonical preferential-trade-agreement model. Every closed form is verified numerically in the replication package.

Notation. Countries $i, j \in \{1, \ldots, N\}$; A is the tariff imposer, B the target, C a bystander. e_{ij} = units of i's currency per unit of j's currency, so a *rise* in e_{ij} is a depreciation of i against j; $e_{ii} = 1$ and $e_{ij} = e_{ik}e_{kj}$. Perturbations: $\nu_j \equiv d\log e_{Aj}$ with $\nu_A = 0$ (A's currency is the numeraire). Tariffs $\tau_{ij} \geq 0$: ad valorem, levied by i on imports from j, revenue rebated lump-sum. CES weights enter shares linearly (share-linear convention). Recurring composites:

$$m \equiv \alpha_T(1-\alpha_D), \qquad \rho_1^* \equiv 1+\alpha_D(\eta-1)-\alpha_T(1-\alpha_D), \qquad D_N \equiv N\alpha_D(\eta-1)+(N-2)\rho+1. \quad (1)$$

1 Environment

1.1 Technology

Each country produces a tradable T_i (only i produces it, all countries consume it) and a non-tradable N_i, from labor alone:

$$Y_{T_i} = A_{T_i}L_{T_i}, \qquad Y_{N_i} = A_{N_i}L_{N_i}, \qquad L_{T_i} + L_{N_i} = L_i. \quad (2)$$

Perfect competition and the normalization $P_{T_i} = 1$ (producer price in i's own currency) give

$$w_i = A_{T_i}, \qquad P_{N_i} = \frac{w_i}{A_{N_i}} = \frac{A_{T_i}}{A_{N_i}}. \quad (3)$$

Wages are constant in own currency; all adjustment runs through exchange rates (Section 2.5).

1.2 Preferences: two-layer nest

$$C_i = C_{T_i}^{\alpha_T}\, C_{N_i}^{1-\alpha_T} \qquad \text{(Cobb–Douglas outer, tradable share } \alpha_T\text{)} \tag{4}$$

$$C_{T_i} = \Big[\, \alpha_D^{1/\eta}\, C_{T_i i}^{\frac{\eta-1}{\eta}} + (1-\alpha_D)^{1/\eta}\, M_i^{\frac{\eta-1}{\eta}} \Big]^{\frac{\eta}{\eta-1}} \qquad \text{(home vs. imports, elasticity } \eta, \text{ home weight } \alpha_D\text{)} \tag{5}$$

$$M_i = \Big[\, \textstyle\sum_{j\neq i} \beta_{ij}^{1/\rho}\, C_{T_j i}^{\frac{\rho-1}{\rho}} \Big]^{\frac{\rho}{\rho-1}} \qquad \text{(across origins, elasticity } \rho, \text{ weights } \textstyle\sum_{j\neq i}\beta_{ij} = 1\text{)} \tag{6}$$

The symmetric benchmark used below imposes more than equal origin weights $\beta_{ij} = 1/(N-1)$: countries also share the preference parameters $(\alpha_T, \alpha_D, \eta, \rho)$ and have equal labor endowments and productivities, so all countries have equal size at the free-trade baseline. Home and foreign tradables remain asymmetric through the home weight α_D. Equal treatment of all tradable origins would be a still stronger restriction — it additionally requires $\rho = \eta$ and $\alpha_D = 1/N$ — and is not imposed. The single-elasticity CES is the special case $\rho = \eta$; Cobb–Douglas substitution between the domestic tradable and the import composite is the special case $\eta = 1$, which fixes the elasticity, not the weight α_D.

1.3 Policy and closure

The consumer price of T_j in i is $p_{ij} = (1+\tau_{ij})\, e_{ij}$ (in i's currency). Nothing requires $\tau_{ii} = 0$: a tax on home tradables ($p_{ii} = 1 + \tau_{ii}$, rebated like the rest) fits the same structure. We set $\tau_{ii} = 0$ throughout, so that τ is a border instrument and $p_{ii} = 1$. Revenue is rebated lump-sum (Section 3.2).

Assumption 1 (Closure)**.** *There are no internationally traded assets: each country's trade balance is zero every period,* $\mathrm{TB}_i = 0$ *for all* i*.*

Assumption 1 closes the model and pins down the $N-1$ independent relative exchange rates.

2 Prices, incomes, and exchange-rate concepts

2.1 Price indices

CES price indices in i's currency, from (5)–(6):

$$P_{M_i} = \Big[\textstyle\sum_{j\neq i} \beta_{ij}\, p_{ij}^{1-\rho}\Big]^{\frac{1}{1-\rho}}, \qquad P_{T_i} = \Big[\alpha_D\, p_{ii}^{1-\eta} + (1-\alpha_D)\, P_{M_i}^{1-\eta}\Big]^{\frac{1}{1-\eta}}, \tag{7}$$

and the consumer price index from (4) (κ_i collects share constants):

$$P_i = \kappa_i\, P_{T_i}^{\alpha_T}\, P_{N_i}^{1-\alpha_T}. \tag{8}$$

2.2 Definitions: the objects to be signed

$$\omega_{ij} \equiv \frac{e_{ij} w_j}{w_i} = \frac{A_{T_j}}{A_{T_i}} e_{ij} \ \propto \ e_{ij} \qquad \text{(relative wage, common currency)} \tag{9}$$

$$\mathrm{ToT}_{ij} \equiv \frac{P_{T_i}}{e_{ij} P_{T_j}} = \frac{1}{e_{ij}} \qquad \text{(bilateral terms of trade, pre-tariff prices)} \tag{10}$$

$$q_{ij} \equiv \frac{e_{ij} P_j}{P_i} \qquad \text{(bilateral real exchange rate)} \tag{11}$$

$$d\nu_i^E \equiv \sum_{j \neq i} w_{ij}\, d\log e_{ij}, \qquad w_{ij} = \frac{f_{ij} + f_{ji}}{\sum_{k \neq i} (f_{ik} + f_{ki})} \qquad \text{(NEER, trade weights)} \tag{12}$$

$$dq_i^E \equiv \sum_{j \neq i} w_{ij}\, d\log q_{ij} \qquad \text{(REER, same weights)} \tag{13}$$

where f_{ij} is the border value of i's imports from j in a common currency — the same flows that enter the trade balances, defined formally in (18) below (the currency choice cancels from the weights). All effective rates are depreciation-positive: $d\nu_i^E > 0$ means i's currency loses value on average. In the symmetric model $w_{ij} = 1/(N-1)$.

2.3 Wage–exchange-rate equivalence

Prices and demands depend on w_j and e_{ij} only through the products $e_{ij} w_j$. The model therefore does not separate relative wage adjustment from nominal exchange-rate adjustment: fixing $w_i = A_{T_i}$ and letting e adjust (as here) is equivalent to fixing e and letting relative wages adjust. Moreover, with one factor and constant returns, producer prices are unit labor costs, so (9)–(10) give

$$d\log e_{ij} \ = \ d\log \omega_{ij} \ = \ -\, d\log \mathrm{ToT}_{ij} : \tag{14}$$

the nominal exchange rate, the common-currency relative wage, and the (inverse) terms of trade move one-for-one. Every result below about e_{ij} is simultaneously a statement about all three.

3 Demands and equilibrium

3.1 Expenditure shares

With income I_i, the share-linear CES demands from (4)–(6) give expenditure shares of income (rows sum to α_T over tradables):

$$s_{ii} = \alpha_T\, \alpha_D \left(\frac{p_{ii}}{P_{T_i}}\right)^{1-\eta}, \tag{15}$$

$$s_{ij} = \alpha_T\, (1 - \alpha_D) \left(\frac{P_{M_i}}{P_{T_i}}\right)^{1-\eta} \beta_{ij} \left(\frac{p_{ij}}{P_{M_i}}\right)^{1-\rho}, \qquad j \neq i. \tag{16}$$

Shares are tariff-inclusive: $s_{ij}I_i$ is spending at consumer prices; $s_{ij}I_i/(1+\tau_{ij})$ is the producer-price (border) value.

3.2 Income

A fraction $\tau_{ij}/(1+\tau_{ij})$ of spending $s_{ij}I_i$ is tariff revenue, rebated:

$$I_i = w_i L_i + \sum_{j\neq i} \frac{\tau_{ij}}{1+\tau_{ij}}\, s_{ij} I_i \quad \Longrightarrow \quad I_i = \frac{w_i L_i}{1 - \sum_{j\neq i} \frac{\tau_{ij}}{1+\tau_{ij}}\, s_{ij}}. \tag{17}$$

The shares depend only on prices, so (17) is a closed form.

3.3 Trade balances and equilibrium

In a common currency, at producer prices (the tariff wedge is a domestic transfer),

$$\mathrm{TB}_i = \underbrace{\sum_{k\neq i} f_{ki}}_{\text{exports}} - \underbrace{\sum_{j\neq i} f_{ij}}_{\text{imports}}, \qquad f_{ij} \equiv \frac{s_{ij}\, I_i}{1+\tau_{ij}} \quad (i\text{'s imports from } j\text{, border value}). \tag{18}$$

By Walras' law $\sum_i \mathrm{TB}_i = 0$: only $N-1$ conditions are independent, matching the $N-1$ free exchange rates. Equilibrium: given $\{\tau_{ij}\}$, the rates $\{e_{Aj}\}_{j\neq A}$ solve $\mathrm{TB}_i = 0$ for $i \neq A$.

4 The general linear system

4.1 Linearization at a balanced free-trade baseline

Evaluate at $\tau_{ij} = 0$ with balanced trade, and normalize baseline incomes. Baseline objects: home share within tradables $h_i = \alpha_D$ (symmetric weights case: general h_i carries through identically), within-import shares $b_{ij} = \beta_{ij}$, income shares $s_{ij} = \alpha_T(1-\alpha_D)b_{ij} = m\, b_{ij}$, and flow values $f_{ij} = s_{ij}I_i$ (from (18) at $\tau = 0$). Balance requires $\sum_k f_{ki} = \sum_j f_{ij}$.

Perturb by tariffs dt_{ij} and collect exchange-rate changes $\nu_j = d\log e_{Aj}$. Log price changes in i's own currency ($dp_{ii} = 0$):

$$dp_{ij} = \nu_j - \nu_i + dt_{ij}. \tag{19}$$

Index changes, from (7):

$$d\log P_{M_i} = \sum_{j\neq i} b_{ij}\, dp_{ij}, \qquad d\log P_{T_i} = (1-\alpha_D)\, d\log P_{M_i}. \tag{20}$$

Share changes, from (15)–(16), using $d\log P_{M_i} - d\log P_{T_i} = \alpha_D\, d\log P_{M_i}$:

$$d\log s_{ii} = -(1-\eta)(1-\alpha_D)\, d\log P_{M_i}, \tag{21}$$

$$d\log s_{ij} = \underbrace{(1-\rho)\big(dp_{ij} - d\log P_{M_i}\big)}_{\text{across origins}} + \underbrace{(1-\eta)\,\alpha_D\, d\log P_{M_i}}_{\text{home vs. imports}}. \tag{22}$$

Income changes in the common currency, from (17) (revenue is first order only for tariffing countries):

$$d \log I_i = \nu_i + \sum_{j \neq i} s_{ij}\, dt_{ij}. \tag{23}$$

Differentiating (18), each flow contributes value $\times$ log change:

$$d\,\mathrm{TB}_i = \sum_{k \neq i} f_{ki}\big(d\log s_{ki} + d\log I_k - dt_{ki}\big) - \sum_{j \neq i} f_{ij}\big(d\log s_{ij} + d\log I_i - dt_{ij}\big). \tag{24}$$

Equations (19)–(24) are linear in the exchange-rate changes and the tariff changes. Stack the free rates in the vector $d\nu \equiv (\nu_j)_{j \neq A} \in \mathbb{R}^{N-1}$ and all directed bilateral tariff changes in $d\boldsymbol{\tau} \equiv (dt_{jk})_{j \neq k}$. Reading the coefficients off (24), the linearized equilibrium conditions for $i \neq A$ are

$$J\, d\nu + G\, d\boldsymbol{\tau} = 0, \qquad J_{il} \equiv \frac{\partial\, d\mathrm{TB}_i}{\partial \nu_l}, \qquad G_{i,(jk)} \equiv \frac{\partial\, d\mathrm{TB}_i}{\partial\, dt_{jk}}\bigg|_{\nu \text{ fixed}}. \tag{25}$$

J is the $(N-1) \times (N-1)$ adjustment mechanism: how trade balances respond to exchange rates. G maps any pattern of tariff changes into the trade-balance disturbances it creates before exchange rates move. A policy experiment is a direction $a = (a_{jk})$ scaled by a single number, $d\boldsymbol{\tau} = a\, d\tau$ (the unilateral tariff of the sections below is $a_{AB} = 1$, all other entries zero; the trade war is $a_{AB} = a_{BA} = 1$), with impact vector

$$F \equiv Ga, \qquad F_i = \frac{\partial\, d\mathrm{TB}_i}{\partial\, d\tau}\bigg|_{\nu = 0}. \tag{26}$$

One sign convention is fixed once and used throughout: F is the *impact effect* of the experiment on the trade balances at unchanged exchange rates, so (25) reads

$$J\, d\nu = -F\, d\tau \quad \Longrightarrow \quad \frac{d\nu}{d\tau} = (-J)^{-1} F = \frac{\mathrm{adj}(-J)\, F}{\det(-J)}, \tag{27}$$

where $\mathrm{adj}(-J)$ is the adjugate (transpose of the cofactor matrix). Different tariff configurations are different directions a applied to the same G: this is the superposition used in Sections 6.5 and 7.4. The entries of J and G are linear in η and ρ with baseline shares as coefficients.

4.2 Properties of J: the multilateral Marshall–Lerner condition

The two-country Marshall–Lerner condition says that a depreciation improves the depreciating country's trade balance. With more than two countries, the single scalar slope is replaced by the Jacobian J, so the condition must be placed on the whole system rather than on one derivative. This subsection builds that condition from the model and shows that it, local stability, and the sign structure of $(-J)^{-1}$ are one notion, not three.

Write $\widetilde{J}$ for the full $N \times N$ Jacobian, $\widetilde{J}_{il} = \partial\, d\mathrm{TB}_i / \partial \nu_l$ for all $i, l \in \{1, \ldots, N\}$, before dropping country A: the J of (25) is $\widetilde{J}$ with row and column A deleted. Two properties of $\widetilde{J}$ are identities of the model, not assumptions:

$$\sum_l \widetilde{J}_{il} = 0 \quad \forall i \qquad \text{(homogeneity: (19) depends only on differences } \nu_j - \nu_i) \tag{28}$$

$$\sum_i \widetilde{J}_{il} = 0 \quad \forall l \qquad \text{(Walras' law: } \textstyle\sum_i \mathrm{TB}_i \equiv 0). \tag{29}$$

Assumption 2 (Gross substitutability)**.** *$\widetilde{J}_{il} \geq 0$ for all $l \neq i$: a depreciation of any country l (a rise in ν_l) does not worsen any other country i's trade balance.*

The condition is on every off-diagonal entry — all country pairs, not only A's partners. In the explicit symmetric systems of Sections 6–7, every off-diagonal entry is a positive multiple of D_N, so Assumption 2 there is exactly $D_N > 0$ and holds for all $\eta \geq 1$.

Combining (28) with Assumption 2 gives the diagonal, and then the reduced system inherits a dominant diagonal:

$$\widetilde{J}_{ii} = -\sum_{l \neq i} \widetilde{J}_{il} \;\leq\; 0, \qquad |J_{ii}| = \sum_{l \neq i,\, A} J_{il} \;+\; \widetilde{J}_{iA} \;\geq\; \sum_{l \neq i,\, A} J_{il}. \tag{30}$$

The first equation is the set of bilateral Marshall–Lerner conditions, one per country: own depreciation improves own balance. The second says each row of $-J$ dominates its off-diagonal sum, strictly in every row i with $\widetilde{J}_{iA} > 0$ (every country whose balance responds to A's rate). So diagonal dominance is not an independent assumption and not merely a restatement of the identities: it follows from homogeneity (28) *plus* gross substitutability, once the numeraire column is dropped. Walras' law (29) delivers the same dominance columnwise.

Result 4.1 (Multilateral Marshall–Lerner condition)**.** *Under Assumption 2, if J is irreducible (a connected trade network) with $\widetilde{J}_{iA} > 0$ for some i, then $-J$ is a nonsingular M-matrix: a matrix with nonpositive off-diagonal entries satisfying the equivalent conditions*

$$\det(-J) > 0 \ \textit{(all principal minors positive)}, \qquad (-J)^{-1} \geq 0 \ \textit{elementwise}, \qquad \operatorname{Re}\lambda(J) < 0. \tag{31}$$

We refer to (31) *as the* multilateral Marshall–Lerner condition*.*

The proof is the standard characterization of irreducibly diagonally dominant Z-matrices; (30) supplies the dominance. Three clarifications of the logic:

- Gross substitutability is a *sufficient primitive*, not the condition itself: Assumption 2 implies (31), but (31) can hold without it. Everything below uses only (31).
- The eigenvalue statement in (31) is local stability of the price-adjustment process $\dot{\nu}_i = k_i\, \mathrm{TB}_i$ with any speeds $k_i > 0$ (the *tatonnement*: a country in surplus sees its currency appreciate). Stability is thus not an extra requirement layered on top of (31) — it is one of its equivalent faces, and M-matrices are stable for every choice of speeds.
- At $N = 2$, all of (31) collapses to the single inequality $J < 0$, whose classical elasticity form $\varepsilon_X + \varepsilon_M > 1$ is derived in Section 5.3.

4.3 The sign rule

With J and F as defined in Section 4.1, the solution $d\nu/d\tau = (-J)^{-1}F$ combines the impact effects through a nonnegative matrix:

Result 4.2 (Sign rule)**.** *Under the multilateral Marshall–Lerner condition* (31)*,*

$$\operatorname{sign}\left(\frac{d\nu_i}{d\tau}\right) = \operatorname{sign}\left(\sum_j \omega_{ij} F_j\right), \qquad \omega_{ij} \equiv \left[(-J)^{-1}\right]_{ij} \geq 0. \tag{32}$$

Every response is a nonnegative combination of the impact effects. If all F_j share one sign, every exchange rate moves the same way; opposite-signed responses require an impact vector with mixed signs.

For the discriminatory tariff studied below, the impact vector is generically mixed, so the sign rule has content. At fixed exchange rates the tariff always worsens the target's balance ($F_B < 0$: B loses export demand to A), while a bystander's balance combines a gain (demand diverted from B, increasing in ρ) with a loss (sales to a poorer B): in the symmetric three-country case $F_C \propto \rho - \rho_1^*$ (Section 6.1), positive exactly when diversion dominates. The thresholds derived below are the points where a weighted impact sum $\sum_j \omega_{ij} F_j$ crosses zero: the ambiguity in bilateral responses comes from the direction of F, never from a failure of the adjustment mechanism J. Because F is linear in ρ and $\operatorname{adj}(-J)$ has degree $N-2$ in ρ, each such sum is a polynomial of degree $N-1$ in ρ: one relevant root at $N = 3$ (Section 6), possibly several sign changes at $N \geq 4$.

5 Two countries

5.1 The slope

At $N = 2$ there is one balance and one rate. Specializing (19)–(24) (details: Section 7 at $N = 2$; ρ is inoperative with a single origin):

$$\boxed{\left.\frac{d \log e_{AB}}{d\tau}\right|_{\tau=0} = -\frac{\rho_1^*}{D_2}, \qquad D_2 = 2\alpha_D(\eta - 1) + 1.} \tag{33}$$

The numerator is the object defined in (1), with the identity

$$\rho_1^* = 1 + \alpha_D(\eta - 1) - \alpha_T(1 - \alpha_D) = \underbrace{\alpha_D \eta}_{\text{home switching}} + \underbrace{(1 - \alpha_D)(1 - \alpha_T)}_{\text{expenditure absorption}} > 0 \tag{34}$$

for all admissible parameters.

Result 5.1 (Conventional wisdom, $N = 2$)**.** *At $N = 2$ the multilateral Marshall–Lerner condition* (31) *reduces to the single inequality $J < 0$, i.e. $D_2 > 0$ — automatic for $\eta \geq 1$, failing only under strong complementarity $\eta < 1 - 1/(2\alpha_D)$, and shown in Section 5.3 to be exactly the textbook condition $\varepsilon_X + \varepsilon_M > 1$. Under it, a unilateral import tariff appreciates the tariffing country's currency unconditionally, since $\rho_1^* > 0$ by* (34)*.*

The sign therefore has two separate ingredients. The Marshall–Lerner condition signs the *denominator*: exchange rates move trade balances in the normal direction. The identity $\rho_1^* > 0$ signs the *numerator*: the trade-balance disturbance the tariff creates. At $N = 2$ both are unambiguous; from $N = 3$ on, only the denominator remains so.

5.2 Exact solution in the Cobb–Douglas case

At $\eta = 1$ shares are constant: $s_{AB} = s_{BA} = m$. From (17), $I_A = w_A L_A / \big[1 - \frac{\tau}{1+\tau} m\big]$, and $\mathrm{TB}_A = 0$ in (18) reads $e\, m\, w_B L_B = m I_A/(1+\tau)$:

$$\boxed{e(\tau) = \frac{w_A L_A}{w_B L_B} \cdot \frac{1}{1 + (1-m)\tau}} \qquad \left.\frac{d \log e}{d\tau}\right|_{\tau=0} = -(1-m) = -\left.\frac{\rho_1^*}{D_2}\right|_{\eta=1}, \tag{35}$$

strictly decreasing in τ globally: the linearized result (33) is exact-in-sign at all tariff levels in the Cobb–Douglas case. Note that $\eta = 1$ pins the expenditure shares at the *weights* $(\alpha_D, 1-\alpha_D)$ but does not restrict the weights themselves: Cobb–Douglas means unit elasticity, not equal shares. Equal home and import spending, $\alpha_D = 1/2$, is a further special case.

5.3 Connection to the textbook Marshall–Lerner condition

This subsection imposes nothing new: it restates the inequality $J < 0$ in the classical elasticity form in which the conventional wisdom is usually asserted, and then shows that in the nested model the elasticity condition and the denominator D_2 are the same object.

Reduced form: export volume $X(p^*)$ with $p^* = 1/e$, import volume $M(p)$ with $p = (1+\tau)e$, elasticities $\varepsilon_X \equiv -X' p^*/X \geq 0$, $\varepsilon_M \equiv -M' p/M \geq 0$. The balance in A's currency at producer prices:

$$\mathrm{TB}_A(e, \tau) = X(1/e) - e\, M\big((1+\tau)e\big). \tag{36}$$

Differentiate at $\tau = 0$, balanced trade ($X = eM$):

$$\frac{\partial \mathrm{TB}_A}{\partial e} = \underbrace{\frac{\varepsilon_X X}{e}}_{\text{export volume}} + \underbrace{\varepsilon_M M}_{\text{import volume}} - \underbrace{M}_{\text{valuation}} = M\big(\varepsilon_X + \varepsilon_M - 1\big). \tag{37}$$

A depreciation improves the balance iff $\varepsilon_X + \varepsilon_M > 1$: the volume responses must offset the one-for-one valuation loss. The tariff response at fixed e:

$$\frac{\partial \mathrm{TB}_A}{\partial \tau} = \frac{\varepsilon_M}{1+\tau}\, eM > 0 \quad \text{whenever } \varepsilon_M > 0. \tag{38}$$

Equilibrium $\mathrm{TB}_A(e(\tau), \tau) = 0$ and the implicit function theorem give

$$\frac{de}{d\tau} = -\frac{\partial \mathrm{TB}_A/\partial \tau}{\partial \mathrm{TB}_A/\partial e} : \tag{39}$$

the tariff creates a surplus at the initial rate (38); the rate must move in the direction that worsens the balance; under Marshall–Lerner that direction is appreciation. The condition is necessary and sufficient because $\operatorname{sign}(de/d\tau) = -\operatorname{sign}(\varepsilon_X + \varepsilon_M - 1)$.

In the nested model the condition is not free-standing. Comparing (37) with the linearized J at $N = 2$ (per unit of import value m):

$$\varepsilon_X + \varepsilon_M - 1 = D_2 = 2\alpha_D(\eta - 1) + 1, \qquad \varepsilon_X = \varepsilon_M = \alpha_D(\eta - 1) + 1 : \tag{40}$$

the denominator D_2 *is* the Marshall–Lerner margin, so (31) at $N = 2$ and $\varepsilon_X + \varepsilon_M > 1$ are the same condition. At $\eta = 1$ (Cobb–Douglas), $\varepsilon_X = \varepsilon_M = 1$ and the margin is exactly 1.

5.4 Quantities

From (9)–(11), with $\nu_B = d\log e_{AB}$:

$$d\log\omega_{AB} = \nu_B, \qquad d\log \mathrm{ToT}_{AB} = -\nu_B, \qquad d\log q_{AB} = (1-2m)\,\nu_B \;-\; m\,d\tau. \tag{41}$$

The real-rate expression is the $N = 2$ case of (68) below: the tariff appreciates A in real terms directly (it raises A's consumer prices at any e), and nominal pass-through is attenuated by the two countries' tradable exposures. A tariff therefore appreciates the currency, improves the terms of trade, raises the relative wage, and appreciates the real exchange rate: at $N = 2$ every definition of "the exchange rate" moves together.

6 Three countries

6.1 The linear system

Set $N = 3$, symmetric weights $\beta_{ij} = 1/2$, equal sizes, and perturb by $d\tau = dt_{AB}$. Baseline shares $s_{ii} = \alpha_T\alpha_D$, $s_{ij} = m/2$. From (19)–(20):

$$d\log P_{M_A} = \tfrac{1}{2}(\nu_B + \nu_C + d\tau), \qquad d\log P_{M_B} = \tfrac{1}{2}\nu_C - \nu_B, \qquad d\log P_{M_C} = \tfrac{1}{2}\nu_B - \nu_C, \tag{42}$$

(own-currency), and $d\log I_A = \frac{m}{2}d\tau$, $d\log I_B = \nu_B$, $d\log I_C = \nu_C$ (common currency). Substituting into (24) for $i = B, C$ and collecting terms in $(\nu_B, \nu_C, d\tau)$ yields, with $D \equiv D_3 = 3\alpha_D(\eta-1)+\rho+1$,

$$J = -\frac{mD}{4}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad F = -\frac{m}{4}\begin{pmatrix} \rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix}. \tag{43}$$

The elasticities enter J only through the scalar D; the matrix part is pure counting. Inverting with $\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)^{-1} = \frac{1}{3}\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$:

$$\begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = (-J)^{-1}F\,d\tau = -\frac{1}{3D}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} \rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix} d\tau = \frac{1}{3D}\begin{pmatrix} -(\rho + 3\rho_1^*) \\ \rho - 3\rho_1^* \end{pmatrix} d\tau. \tag{44}$$

Result 6.1 (Threshold)**.** *At the symmetric point the multilateral Marshall–Lerner condition* (31) *reduces to $D > 0$, automatic for $\eta \geq 1$ and $\rho > 0$. Under it,*

$$\frac{d\log e_{AB}}{d\tau} = -\frac{\rho + 3\rho_1^*}{3D} < 0, \qquad \boxed{\frac{d\log e_{AC}}{d\tau} = \frac{\rho - 3\rho_1^*}{3D}} \qquad \frac{d\log e_{BC}}{d\tau} = \frac{2\rho}{3D} > 0. \tag{45}$$

A appreciates against the target B for all parameters; A depreciates against the untariffed C iff $\rho > \rho_3^* \equiv 3\rho_1^*$; B depreciates against C for all parameters.

In the sign rule (32), $(-J)^{-1} = \frac{4}{3mD}\left(\begin{smallmatrix}2&1\\1&2\end{smallmatrix}\right)$ and

$$\frac{d\nu_C}{d\tau} \propto F_B + 2F_C \propto \rho - 3\rho_1^* : \tag{46}$$

the threshold is the point where C's own diversion gain ($F_C > 0$ iff $\rho > \rho_1^*$) outweighs its exposure to B's loss ($F_B < 0$). A higher ρ makes foreign suppliers easier to substitute across, so more of the expenditure displaced from the tariffed B is redirected toward C; the response against C changes sign only when that diversion becomes strong enough.

6.2 Impossibility under a single elasticity

Impose $\rho = \eta = \sigma$. Then

$$\rho_3^* - \sigma = 3\rho_1^*\big|_{\eta=\sigma} - \sigma = \sigma(3\alpha_D - 1) + 3(1-\alpha_D)(1-\alpha_T). \tag{47}$$

Result 6.2 (Impossibility)**.** *If $\alpha_D \geq 1/3$, then $\rho_3^* > \sigma$ for every $\sigma > 0$: with one elasticity, the reversal against C is unreachable. The bound involves only α_D; α_T is unrestricted. As an illustration, at the point $\alpha_D = 1/3$, $\alpha_T = 3/4$ — equal Cobb–Douglas weights $1/4$ on each of the four goods (three tradables and the non-tradable), the paper's original calibration, not a requirement of the result — $\rho_3^* - \sigma = \frac{1}{2}$, $D = 2\sigma$, and*

$$\frac{d\log e_{AC}}{d\tau} = \frac{\sigma - \rho_3^*}{3D} = -\frac{1}{12\sigma} \longrightarrow 0^-. \tag{48}$$

6.3 Bilateral versus effective: the NEER

Decompose the solution (44) into an aggregate and a relative component:

$$\underbrace{\frac{\nu_B + \nu_C}{2}}_{\text{NEER of } A} = -\frac{\rho_1^*}{D}\,d\tau, \qquad \underbrace{\nu_B - \nu_C}_{\text{relative}} = -\frac{2\rho}{3D}\,d\tau. \tag{49}$$

The cross-origin elasticity ρ cancels from the aggregate numerator and fully controls the relative component. The denominator D still contains ρ, so cross-origin substitution affects the magnitude of the NEER response but not its sign: economically, ρ governs how the exchange-rate adjustment is distributed across foreign partners.

Result 6.3 (NEER preservation)**.** *With symmetric weights, $d\nu_A^E/d\tau = -\rho_1^*/D < 0$ for all ρ: a unilateral tariff always appreciates the tariffing country's effective exchange rate, even when $\rho > \rho_3^*$*

reverses the bilateral rate against C. The conventional wisdom fails bilaterally but survives in aggregate, and by (34) *the aggregate sign is unconditional.*

Where the conditions sit empirically. Given (31), $D > 0$ and $\rho_1^* > 0$ hold at every admissible calibration, so the signs of e_{AB}, e_{BC}, and the NEER in this section are parameter-free. The one genuinely ambiguous object is the bilateral rate against the bystander: at the benchmark, $\rho_3^* = 3.96$ sits inside the range of estimated cross-origin trade elasticities (roughly 2–8: macro estimates at the low end, trade-level estimates at the high end), so either sign of $d\log e_{AC}$ is empirically plausible. By contrast the war threshold below, $\rho_1^* = 1.32$, lies beneath essentially all estimates, so the war predictions are clear-cut.

6.4 Real exchange rates

From (11) and (8), $d\log q_{Aj} = \nu_j + \alpha_T\big(d\log P_{T_j}^{\text{own}} - d\log P_{T_A}^{\text{own}}\big)$ with own-currency indices (20). Both bilateral real rates take one form (the $N = 3$ case of (68)):

$$d\log q_{Aj} = \Big(1 - \tfrac{3m}{2}\Big)\,\nu_j \;-\; \tfrac{m}{2}\,d\tau, \qquad j = B, C. \tag{50}$$

Setting $d\log q_{AC} = 0$ with ν_C from (44): $(2-3m)(\rho - 3\rho_1^*) = 3mD$, linear in ρ, so

$$\boxed{\rho_q^* = \frac{(2-3m)\,3\rho_1^* + 3m\big[\,3\alpha_D(\eta-1)+1\,\big]}{2(1-3m)}} \qquad (\rho_q^* > \rho_3^* \text{ for } m \in (0, \tfrac{1}{3})). \tag{51}$$

At the benchmark ($\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$, $m = 0.08$): $\rho_3^* = 3.96$, $\rho_q^* = 4.93$. Averaging (50) over j gives the REER:

$$\frac{dq_A^E}{d\tau} = -\Big(1 - \tfrac{3m}{2}\Big)\frac{\rho_1^*}{D} - \frac{m}{2} \;<\; 0 \qquad \text{for all } \rho\ (m < \tfrac{2}{3}) : \tag{52}$$

real effective appreciation is unconditional and strictly stronger than nominal, because the tariff also raises A's consumer prices directly.

6.5 Tariff configurations by superposition

Responses are linear in the tariff vector: any configuration is a sum of directed bilateral shocks, with mirror shocks obtained by relabeling (44).

$$\text{broad tariff (on } B \text{ and } C\text{):} \qquad \frac{d\log e_{Aj}}{d\tau} = \nu_B + \nu_C = -\frac{2\rho_1^*}{D} < 0, \quad j = B, C, \tag{53}$$

$$\text{symmetric war (with } B\text{):} \qquad \left.\frac{d\log e_{AB}}{d\tau}\right|^{w} = 0, \qquad \left.\frac{d\log e_{AC}}{d\tau}\right|^{w} = \frac{\rho - \rho_1^*}{D}. \tag{54}$$

In (53), ρ drops out entirely: uniform protection is non-discriminatory, so there is no cross-origin margin and every bilateral rate (hence the NEER) appreciates. In (54), the belligerents' bilateral component cancels and the bystander margin decides everything:

$$\left.\frac{d\nu_A^E}{d\tau}\right|^w = \frac{1}{2}\,\frac{\rho-\rho_1^*}{D} > 0 \quad\iff\quad \rho > \rho_1^*. \tag{55}$$

Result 6.4 (Configuration dependence)**.** *The unilateral tariff appreciates the NEER for all ρ; the reciprocal war depreciates both belligerents' NEERs iff $\rho > \rho_1^*$. Retaliation, not protection itself, is what can weaken the tariffing country's currency in aggregate.*

7 General symmetric N

7.1 The reduced system

Symmetric weights $b_{ij} = 1/(N-1)$, equal sizes, $d\tau = dt_{AB}$; the $N-2$ bystanders are identical with common ν_C. From (19)–(20):

$$d\log P_{M_A} = \frac{\nu_B + d\tau + (N-2)\nu_C}{N-1}, \qquad d\log P_{M_B} = \frac{(N-2)\nu_C}{N-1} - \nu_B, \qquad d\log P_{M_C} = \frac{\nu_B - 2\nu_C}{N-1}, \tag{56}$$

and $d\log I_A = \frac{m}{N-1}d\tau$, $d\log I_B = \nu_B$, $d\log I_C = \nu_C$. Substituting into (24) for B and a representative bystander and collecting terms (each row scaled by $(N-1)/m$; verified symbolically):

$$D_N \begin{pmatrix} -(N-1) & N-2 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = \begin{pmatrix} (N-2)\rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix} d\tau, \qquad D_N = N\alpha_D(\eta-1) + (N-2)\rho + 1. \tag{57}$$

As at $N = 3$, the elasticities enter the adjustment matrix only through the scalar D_N; the matrix is pure counting, with determinant N. Inverting:

$$\frac{d\log e_{AB}}{d\tau} = -\,\frac{N\rho_1^* + (N-2)\rho}{N\,D_N} \;<\; 0, \tag{58}$$

$$\frac{d\log e_{AC}}{d\tau} = \frac{\rho - N\rho_1^*}{N\,D_N}, \tag{59}$$

$$\frac{d\log e_{BC}}{d\tau} = \frac{(N-1)\,\rho}{N\,D_N} \;>\; 0. \tag{60}$$

At $N = 3$ these are (45); at $N = 2$, (58) is (33) (with ρ absent, $D_2 = 2\alpha_D(\eta-1)+1$).

Result 7.1 (N-country threshold)**.** *A depreciates against each untariffed bystander iff*

$$\boxed{\rho > \rho_N^* = N\rho_1^*} \tag{61}$$

The threshold is linear in N: diverted expenditure splits across $N-2$ bystanders, each receiving weight $1/(N-1)$, so a larger ρ is needed to reverse any one bilateral rate. This is a statement about how the adjustment is distributed across partners, not about the aggregate value of A's currency.

7.2 The NEER: dimension- and ρ-invariant sign

Averaging with symmetric weights:

$$\frac{d\nu_A^E}{d\tau} = \frac{\nu_B + (N-2)\nu_C}{N-1} = \frac{-\big[N\rho_1^* + (N-2)\rho\big] + (N-2)\big[\rho - N\rho_1^*\big]}{(N-1)\,N\,D_N} = \boxed{-\frac{\rho_1^*}{D_N}} \tag{62}$$

The ρ terms cancel exactly, and N affects only the magnitude through D_N. The companion relative component is pure ρ:

$$\nu_B - \nu_C = -\frac{(N-1)\rho}{N D_N}\,d\tau. \tag{63}$$

Result 7.2 (Aggregation)**.** *In the symmetric model, for every $N \geq 2$ and every ρ, a unilateral tariff appreciates the tariffing country's NEER: $d\nu_A^E/d\tau = -\rho_1^*/D_N < 0$ by* (34)*. Cross-origin substitution reallocates the adjustment across partners* (63) *and can reverse individual bilateral rates* (59)*, but under symmetry it cannot reverse the NEER. This is the precise sense in which the conventional two-country appreciation result survives multilateralization.*

7.3 Impossibility at general N

With $\rho = \eta = \sigma$:

$$\rho_N^* - \sigma = \sigma(N\alpha_D - 1) + N(1-\alpha_D)(1-\alpha_T) \;>\; 0 \quad \text{whenever } \alpha_D \geq 1/N. \tag{64}$$

The home-bias bound weakens with N: any calibration with home share at least $1/N$ keeps the bilateral reversal invisible to single-elasticity models.

7.4 Configurations

By superposition, as in Section 6.5:

$$\text{broad tariff (on all } N-1 \text{ partners):} \qquad \frac{d\log e_{Aj}}{d\tau} = -\frac{(N-1)\rho_1^*}{D_N} \quad \forall j \;=\; \frac{d\nu_A^E}{d\tau}, \tag{65}$$

$$\text{symmetric war with } B: \qquad \left.\frac{d\log e_{AB}}{d\tau}\right|^w = 0, \qquad \left.\frac{d\log e_{AC}}{d\tau}\right|^w = \frac{\rho-\rho_1^*}{D_N}, \tag{66}$$

and the war NEER, $\frac{N-2}{N-1}\,\frac{\rho-\rho_1^*}{D_N}$, has the sign of $\rho - \rho_1^*$.

Result 7.3 (War threshold is independent of N)**.** *In a symmetric war, every bystander appreciates against both belligerents — and both belligerents' NEERs depreciate — iff*

$$\boxed{\rho > \rho_1^* = \frac{\rho_N^*}{N}} \tag{67}$$

for any N. With e_{AB} pinned at zero by symmetry, each bystander's equation decouples and its sign is its own impact effect, which does not dilute with N. At $\eta = 1$, $\rho_1^* = 1 - m < 1$: any $\rho \geq 1$ suffices.

7.5 Real rates at general N

Both bilateral real rates take one form (verified symbolically; contains (41) and (50)):

$$d\log q_{Aj} = \Big(1 - \frac{mN}{N-1}\Big)\,\nu_j \;-\; \frac{m}{N-1}\,d\tau, \qquad j = B, C. \tag{68}$$

Setting $d\log q_{AC} = 0$ with (59) gives the real bilateral threshold (for $m < 1/N$):

$$\boxed{\rho_q^*(N) = \frac{\big[(N-1) - mN\big]\,N\rho_1^* + mN\big[N\alpha_D(\eta-1)+1\big]}{(N-1)(1-mN)}} \tag{69}$$

($\rho_q^*(3) = 4.93$, $\rho_q^*(4) = 7.34$, $\rho_q^*(5) = 10.40$ at the benchmark), and averaging (68) gives the REER:

$$\frac{dq_A^E}{d\tau} = -\Big(1 - \frac{mN}{N-1}\Big)\frac{\rho_1^*}{D_N} - \frac{m}{N-1} \;<\; 0 \qquad \text{for all } \rho, N \ \big(m < \tfrac{N-1}{N}\big). \tag{70}$$

7.6 Correspondence across N

object	$N=2$	$N=3$	general N
ML condition (31)	$\varepsilon_X + \varepsilon_M - 1 = D_2 > 0$	$D_3 > 0$	$-J$ a nonsingular M-matrix
sufficient primitive	$\eta \geq 1$	$\eta \geq 1$	gross substitutability (Assumption 2)
sign rule	one impact, unconditional	weights $\frac{1}{3}\left(\begin{smallmatrix}2 & 1\\ 1 & 2\end{smallmatrix}\right)$	sign $\big(\sum_j \omega_{ij}F_j\big)$, $\omega \geq 0$
bilateral threshold	none (no bystander)	$3\rho_1^*$	$N\rho_1^*$ (symmetric); degree-$(N-1)$ roots in general
real bilateral threshold	none	$\rho_q^*(3)$ (51)	$\rho_q^*(N)$ (69)
NEER (unilateral)	$-\rho_1^*/D_2$	$-\rho_1^*/D_3$	$-\rho_1^*/D_N$: sign invariant
war threshold	—	ρ_1^*	ρ_1^*, independent of N

8 Asymmetry

Asymmetry changes the weights placed on the same channels rather than introducing a new channel: the questions are where the bilateral threshold moves, and whether the aggregate NEER result survives.

8.1 General baselines

Nothing in Section 4 used symmetry: at any balanced baseline, the system (19)–(24) holds with the observed shares (b_{ij}, h_i) and incomes y_i as coefficients. The paper's appendix solves it for $N = 3$

at a general baseline (unequal shares and sizes, existing tariffs). The bilateral slope against the bystander then has denominator $\det(-J) > 0$ and numerator quadratic in ρ:

$$\operatorname{sign} \frac{d \log e_{AC}}{d\tau} = \operatorname{sign} \mathcal{N}(\rho), \qquad \mathcal{N}(\rho) = c_2\, \rho^2 + c_1\, \rho + c_0, \qquad c_2 = b_{AB}\, b_{AC}\, b_{CA}\, b_{CB}\, (1-h_A)(1-h_C)\, y_A\, y_C. \tag{71}$$

Two consequences:

- $c_2 > 0$ whenever A and C each buy from both foreign origins (every share in the product is positive). Then $\mathcal{N}(\rho) > 0$ for ρ large enough: sufficiently strong cross-origin substitution reverses e_{AC} at *any* such baseline. Asymmetry moves the threshold; it cannot remove the reversal region.
- The threshold itself is the largest root of $\mathcal{N}(\rho) = 0$. At the symmetric free-trade baseline that root is $3\rho_1^*$, recovering (45).

The remaining coefficients c_1, c_0 depend on the full pattern of bilateral shares and sizes, and their interpretation is the reweighting of the symmetric channels: a country well positioned to replace B in A's import basket receives more of the diversion effect, while a country heavily exposed to the fall in B's income receives more of the offset. The sign of a bilateral response therefore depends on the trade network as well as on (η, ρ), and no single universal threshold exists away from symmetry.

8.2 Asymmetric exposure to the target: the κ parameterization

To compare asymmetry across N, let the target B carry a fixed multiple κ of the symmetric bilateral share, with the weight matrix kept symmetric (hence balanced):

$$b_{AB} = b_{BA} = \frac{\kappa}{N-1}, \qquad b_{AC} = b_{CA} = b_{BC} = b_{CB} = \frac{1 - \frac{\kappa}{N-1}}{N-2}, \qquad b_{CC'} = \frac{1 - 2\, b_{CA}}{N-3}. \tag{72}$$

$\kappa = 1$ is exact symmetry; $\kappa \in (0, N-1]$ scales the target's importance to A while remaining comparable as N changes. The bystanders are still interchangeable, so the reduced system again has two unknowns (ν_B, ν_C), and it solves in closed form (symbolically; replication package): with NEER weights $w_{Aj} = b_{Aj}$, the response $d\nu_A^E/d\tau$ is an explicit rational function of (ρ, N, κ) that collapses to $-\rho_1^*/D_N$ exactly at $\kappa = 1$. Its numerator is $-\kappa\, \rho_1^*$ times a quadratic $Q(\rho)$ with leading coefficient proportional to $(\kappa - 1)(N - 1 - \kappa)$.

Result 8.1 (Asymmetric NEER)**.** *Holding κ fixed as $N \to \infty$,*

$$N\, \frac{d\nu_A^E}{d\tau} \;\longrightarrow\; -\kappa\, \frac{\rho_1^*}{\rho + \alpha_D(\eta - 1)} \;=\; -\kappa \lim_{N\to\infty} \frac{N \rho_1^*}{D_N} : \tag{73}$$

to leading order, asymmetric exposure scales the symmetric NEER response by the target's relative importance κ and preserves its sign.

At finite N the sign of the exact solution follows from Q, and the direction of the asymmetry matters:

- $\kappa \geq 1$ (a disproportionately *large* target): the coefficients of Q are positive for large N (the leading coefficient by $(\kappa-1)(N-1-\kappa) \geq 0$, the others with leading terms N^2 and $N^2\alpha_D(\eta-1)$), so $d\nu_A^E/d\tau < 0$ for every ρ: effective appreciation survives.
- $\kappa < 1$ (a disproportionately *small* target): Q has a single positive root,

$$\rho_E^*(\kappa, N) = \frac{N\rho_1^*}{1-\kappa}\left(1 + O(N^{-1})\right), \tag{74}$$

  beyond which the NEER *depreciates*: with little weight on the target, the depreciations against the many bystanders outweigh the appreciation against B in the average. As $\kappa \to 0$ the flip point approaches the bilateral threshold $N\rho_1^*$ — with a weightless target, the NEER simply *is* the bystander rates.

The exact cancellation of (62) is therefore one-sided. Tariffs on disproportionately large partners preserve the conventional aggregate sign for every ρ; tariffs on minor partners can reverse it, but only at ρ of order N — far above the war threshold ρ_1^* and, at moderate N, above estimated elasticities ($\rho_E^* \approx 22$ at $N = 10$, $\kappa = 0.5$, benchmark parameters).

9 Canonical PTA model: competing exporters

9.1 Setup

Countries A (importer), B and C (exporters). Two goods: x and numeraire y. Endowments: A holds only y; B and C hold x (and possibly y). Quasilinear preferences in each country,

$$U_i = u_i(x_i) + y_i, \qquad u_i' > 0,\ u_i'' < 0. \tag{75}$$

A levies specific tariffs t_B, t_C on imports of x from B and C. Let q_j be the price received by exporter j (the world price of its good) and p the domestic price of x in A. Both origins sell in A at the same domestic price:

$$p = q_B + t_B = q_C + t_C. \tag{76}$$

9.2 Demands and market clearing

From $u_i'(x_i) =$ (local price of x):

$$M_A(p) = x_A^d(p), \qquad E_j(q_j) = \omega_j - x_j^d(q_j), \quad j = B, C, \tag{77}$$

with $M_A' < 0$ and $E_j' > 0$. Market clearing for x:

$$M_A(p) = E_B(p - t_B) + E_C(p - t_C). \tag{78}$$

Equation (78) determines p given (t_B, t_C); then q_B, q_C follow from (76).

9.3 Comparative statics of a discriminatory tariff

Totally differentiate (78) with respect to t_B, holding t_C:

$$M'_A \frac{dp}{dt_B} = E'_B\Big(\frac{dp}{dt_B} - 1\Big) + E'_C \frac{dp}{dt_B} \quad \Longrightarrow \quad \frac{dp}{dt_B} = \frac{E'_B}{E'_B + E'_C - M'_A} \in (0,1). \tag{79}$$

Therefore

$$\frac{dq_B}{dt_B} = -\frac{E'_C - M'_A}{E'_B + E'_C - M'_A} \ < \ 0, \qquad \frac{dq_C}{dt_B} = \frac{dp}{dt_B} \ > \ 0. \tag{80}$$

Result 9.1 (Viner–Mundell)**.** *A discriminatory tariff on B worsens B's terms of trade and* improves *C's terms of trade. Equivalently, a preferential tariff cut toward B (lower t_B) improves B's and worsens C's terms of trade.*

9.4 Why the canonical PTA result is sharper

An improvement in C's terms of trade is a rise in e_{AC} (10). The canonical model signs it unconditionally from $E'_j > 0$ and $M'_A < 0$; the nested model of Section 6 requires $\rho > \rho_3^* = 3\rho_1^*$. The difference is a set of economic margins that the canonical environment removes. Read the threshold as the balance of three forces (benchmark values $\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$):

$$\rho_3^* = 3\rho_1^* = \underbrace{3}_{B\text{–}C\text{ linkage}} + \underbrace{3\alpha_D(\eta-1)}_{\text{home switching}} - \underbrace{3\alpha_T(1-\alpha_D)}_{\text{openness credit}} = 3.0 + 1.2 - 0.24 = 3.96. \tag{81}$$

First, B and C trade with each other in the nested model. When A tariffs B, the income lost in B cuts B's purchases from C, and the change in relative prices induces C to substitute toward B's cheapened goods. Both effects work against the simple idea that expenditure displaced from B must move toward C; they are the source of the first term.

Second, consumers in A can switch toward the domestic tradable rather than toward C. This home-switching channel strengthens with η and with the home share α_D, raising the ρ required for diversion toward C to dominate: the second term.

Third, openness works the other way. The import share $m = \alpha_T(1-\alpha_D)$ measures how much foreign expenditure the tariff can reallocate; a larger share lowers the threshold. This is the credit term.

The decomposition can be checked by recomputing the threshold with each margin removed (severing trade changes the baseline, so the pieces are approximately additive):

restriction	threshold ρ_3^*	piece removed
none (benchmark)	3.96	—
no B–C trade ($\beta_{BC} = \beta_{CB} = 0$)	1.05	the linkage claim
no home tradables ($\alpha_D = 0$)	1.80	the home-switching claim
no non-tradables ($\alpha_T = 1$)	3.60	credit maximized (small alone)
$\alpha_D = 0$ and $\alpha_T = 1$	none: reversal for all $\rho > 0$	
no B–C trade and $\alpha_D = 0$	none: reversal for all $\rho > 0$	

Three readings:

- The largest piece is the B–C linkage: severing it alone moves the threshold from 3.96 to 1.05.
- Non-tradables are not an independent force: with the linkage severed and $\alpha_D = 0$, no threshold exists at any α_T. The non-tradable share enters only through the credit.
- Two routes reach $\rho_3^* \leq 0$: $\{\alpha_D = 0,\ \alpha_T = 1\}$ or $\{\alpha_D = 0,\ \beta_{BC} = \beta_{CB} = 0\}$. The canonical model takes both, adds quasilinearity, and in the Viner case sets $\rho = \infty$.

The canonical model removes most of these forces at once. B and C are competing exporters rather than full trading partners, so there is no B–C demand linkage. A does not produce the imported good, so displaced expenditure cannot move to a domestic variety. Quasilinearity removes the income effects. What remains is reallocation between the two foreign suppliers, signed directly by downward-sloping import demand and upward-sloping export supply.

Result 9.2. *The canonical PTA model is the restriction of the three-country model to a subspace of structures on which $\rho_3^* \leq 0$: the unconditional Viner–Mundell sign is the threshold condition of Section 6 evaluated where it cannot bind. The nested model does not overturn the Viner–Mundell mechanism — it embeds it among the margins above, and $\rho > \rho_3^*$ states when substitution toward the untreated exporter dominates them.*

The analytical literature maps onto the table:

- competing exporters (Bagwell–Staiger 1999; Ornelas 2005): $\beta_{BC} = \beta_{CB} = 0$, quasilinear;
- symmetric endowment blocs (Krugman 1991; Bond–Syropoulos 1996): $\alpha_D \approx 0$, $\alpha_T = 1$, one elasticity;
- Viner: $\rho = \infty$;
- Kemp–Wan 1976: none of these restrictions, and no sign claims.

The applied literature sits on the other side of the same divide. Large simulation models — computable general equilibrium models of trade agreements in the GTAP tradition, and modern quantitative trade models — contain every force in (81): they use a two-level import nest like (5)–(6), full trade matrices, non-tradables, and income effects. In their simulations, a discriminatory tariff sometimes improves and sometimes worsens the excluded country's terms of trade, depending on the configuration. That is what crossing a threshold looks like when the threshold has not been written down: the models produce both signs, but no statement of when each occurs.

The division of labor is then simple. The analytical PTA models prove the sign by deleting the opposing forces. The applied models keep all the forces but report numbers, not conditions. The condition itself is $\rho > 3\rho_1^*$ — and the same three forces run through every result in these notes: dilution across bystanders makes the bilateral threshold $N\rho_1^*$, aggregation across them restores the conventional NEER sign, and a symmetric war cancels the belligerents' bilateral component so that the diversion channel decides even the effective rate. All are manifestations of one distinction: aggregate home-versus-foreign adjustment versus reallocation across foreign suppliers.